

A scattered-target criterion for compatible discretization and faithful discretization of FDI C^* -algebras

Substantial partial result for arXiv:1607.03376

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Status. Likely valid substantial partial result. It gives new positive classes for Question 2.6 of the source, but does not settle the question for arbitrary C^* -algebras.

1 Source question

For a commutative unital C^* -algebra $C = C(X)$, write $L(C) = \ell^\infty(X)$. The source constructs a universal functorial discretization

$$\eta_A : A \longrightarrow F(A) = \operatorname{colim}_{C \subseteq A \text{ commutative}} A *_C L(C).$$

Question 2.6, on source PDF page 5, asks whether η_A is injective or faithful for every A , equivalently whether every A has an injective or faithful compatible C^* -discretization [1].

Whereas the ‘incompatible’ discretization of Proposition 2.3 is faithful, it is not clear whether the natural C^* -discretizations $A \rightarrow F(A)$ of the last theorem are faithful or even injective. Abstract nonsense alone does not answer this question.

Question 2.6. *Is the universal functorial C^* -discretization $\eta_A : A \rightarrow F(A)$ of Theorem 2.5 injective or faithful for every C^* -algebra A ? Equivalently, does every C^* -algebra have an injective or faithful compatible C^* -discretization?*

All algebras and homomorphisms below are unital, as in the source. We call a C^* -algebra scattered when every commutative C^* -subalgebra has scattered Gelfand spectrum. We use the standard fact that, for a scattered compact space X , every Radon measure is atomic and hence

$$C(X)^{**} \cong \ell^\infty(X) \tag{1}$$

canonically as von Neumann algebras.

2 A general criterion

Theorem 1 (scattered-target criterion). *Let A be a unital C^* -algebra and let $\{\pi_i : A \rightarrow B_i\}_{i \in I}$ be unital $*$ -homomorphisms into scattered C^* -algebras. Put $M = \prod_{i \in I} B_i^{**}$ and*

$$\Phi : A \longrightarrow M, \quad \Phi(a) = (\pi_i(a))_i.$$

Then Φ has canonical normal context extensions

$$\Phi_C : \ell^\infty(\operatorname{Spec} C) \longrightarrow M$$

that make it a compatible W^ -discretization. Moreover:*

- (i) Φ is injective if and only if the family (π_i) separates the points of A ;
- (ii) Φ_C is injective if and only if

$$\text{Spec } C = \bigcup_{i \in I} r_{i,C}(\text{Spec}(\pi_i(C))), \quad (2)$$

where $r_{i,C}(y) = y \circ \pi_i|_C$.

Consequently separation together with (2) for every commutative $C \subseteq A$ gives a faithful compatible W^* -discretization.

Proof. Fix a commutative $C \subseteq A$, set $X = \text{Spec } C$ and $Y_i = \text{Spec}(\pi_i(C))$. Since $\pi_i(C)$ is a commutative subalgebra of the scattered algebra B_i , it is scattered. Thus (1) identifies

$$\ell^\infty(Y_i) \cong \pi_i(C)^{**}.$$

The quotient map $C \rightarrow \pi_i(C)$ is Gelfand dual to the injection $r_{i,C} : Y_i \rightarrow X$. Define the i th coordinate of Φ_C by

$$\ell^\infty(X) \xrightarrow{f \mapsto f \circ r_{i,C}} \ell^\infty(Y_i) \cong \pi_i(C)^{**} \xrightarrow{(\pi_i(C) \hookrightarrow B_i)^{**}} B_i^{**}. \quad (3)$$

Every arrow is a normal unital $*$ -homomorphism. On $C = C(X)$ this is exactly $\pi_i|_C$, so Φ_C extends $\Phi|_C$.

If $C \subseteq D$, the four spectral restriction maps satisfy

$$r_{i,C} \circ (\text{Spec}(\pi_i(D)) \rightarrow \text{Spec}(\pi_i(C))) = (\text{Spec } D \rightarrow \text{Spec } C) \circ r_{i,D}.$$

Pullback of bounded functions and bidual functoriality therefore show, one coordinate at a time, that Φ_C factors through Φ_D by the source's induced map $L(C) \rightarrow L(D)$. This is compatibility.

Part (i) follows immediately from the product definition. For (ii), the kernel of the i th restriction in (3) consists of the bounded functions vanishing on $r_{i,C}(Y_i)$. Hence

$$\ker \Phi_C = \{f \in \ell^\infty(X) : f = 0 \text{ on } \bigcup_i r_{i,C}(Y_i)\}.$$

This kernel is zero precisely when the union is all of X : if a point x is missing, its characteristic function δ_x is a nonzero kernel element. $\square$

3 Two positive classes

Corollary 2 (scattered algebras). *If A is scattered, then $A \rightarrow A^{**}$ is a faithful compatible W^* -discretization. In particular, the source's universal $\eta_A : A \rightarrow F(A)$ is faithful.*

Proof. Apply Theorem 1 to the singleton family $\pi = \text{id}_A$. Separation and spectral coverage are tautological. The last assertion follows from universality: the compatible discretization factors as $A \xrightarrow{\eta_A} F(A) \rightarrow A^{**}$. Likewise, each injective context map factors through the corresponding canonical map $L(C) \rightarrow F(A)$, forcing all those canonical maps, as well as η_A , to be injective. $\square$

Recall that an FDI algebra is one whose irreducible representations are all finite-dimensional. For unital algebras this is equivalent to being liminal, or CCR: an irreducible image is $K(H)$, and its unit forces H to be finite dimensional.

Corollary 3 (FDI/CCR algebras). *Every unital FDI (equivalently unital CCR) C^* -algebra has a faithful compatible atomic W^* -discretization. Consequently its universal map $A \rightarrow F(A)$ is faithful.*

Proof. Take one representative π of every irreducible representation. Its image is a full matrix algebra, hence scattered, and irreducible representations separate the points of A .

It remains to prove spectral coverage. Let $C \subseteq A$ be commutative and $x \in \text{Spec } C$. The pure state x of C has a pure-state extension ω to A . Its GNS representation $(\pi_\omega, H_\omega, \xi_\omega)$ is irreducible and therefore finite-dimensional. For every $c \in C$,

$$\|(\pi_\omega(c) - x(c))\xi_\omega\|^2 = \omega((c - x(c))^*(c - x(c))) = 0.$$

Thus ξ_ω is a joint eigenvector for $\pi_\omega(C)$ with character x . Equivalently, x belongs to the image of $\text{Spec}(\pi_\omega(C)) \rightarrow \text{Spec } C$. Since x was arbitrary, (2) holds. Theorem 1 and the same universal factorization argument as in Corollary 2 finish the proof. $\square$

This genuinely exceeds the source’s subhomogeneous result: the unital algebra

$$\left(\bigoplus_{n \geq 1}^{c_0} C([0, 1], M_n) \right)^\sim$$

is FDI, but its irreducible dimensions are unbounded, so it is not subhomogeneous. It also has commutative subalgebras with non-scattered spectra, so Corollaries 2 and 3 provide distinct routes.

4 Why the full question remains

Eight focused attempts are recorded with the packet. Two obstructions prevent promotion to a full answer.

First, although every individual pushout $A *_C L(C)$ contains A faithfully, the compatibility colimit is indexed by the nonfiltered poset of commutative contexts. Context diamonds produce latching maps for which the iterated amalgamated-free-product embedding argument used by the source’s incompatible construction does not apply.

Second, the source’s relatively diffuse pairs force the atomic projections from two context extensions to be mutually orthogonal. In an AW^* -target, arbitrary projection suprema recover both units and give a contradiction. An arbitrary C^* -target has neither those suprema nor normality of maps out of ℓ^∞ , so the same argument does not force a nonzero element of A into the kernel. Passing to the bidual does not repair this: the extended context maps need not agree there as normal maps.

The successful criterion also has a sharp reach obstruction. Simple purely infinite algebras admit no nonzero maps into scattered algebras: a nonzero map would be injective, while subalgebras of scattered algebras are scattered. Thus a full solution requires an idea beyond scattered targets.

5 Verification and novelty

The accompanying finite-set script checks the contravariant spectral square, the exact coverage criterion, and the missing-point characteristic-function kernel witness. These computations audit the combinatorial core; they do not replace the operator-algebra proof.

A bounded novelty search on 12–13 August 2026 used arXiv:1607.03376, the exact question wording, and combinations of “universal functorial C^* -discretization”, “compatible”, “FDI”, “CCR”, “liminal”, and “scattered”. It found the source and its earlier AW^* -obstruction paper, but no explicit statement of Theorem 1 or Corollary 3. Novelty confidence is moderate pending expert citation review.

Human-review recommendation. Check the normal identification (1), the spectral naturality square used for compatibility, and the pure-state extension step in Corollary 3. These are the only substantive external standard facts.

References

- [1] C. Heunen and M. L. Reyes, *Discretization of C^* -algebras*, arXiv:1607.03376v2 (2017), Question 2.6, PDF p. 5.